AgriAssist: Multimodal Time-Series-Based Crop Pest Outbreak Prediction

Phase 2 Interim Report on
Deep Learning Project [ICT-4442]

Submitted By

Nihar P Shetty (Reg No: 230911027)
Pranav Prabhu (Reg No: 230911414)
Vineet Pai (Reg No: 230911450)
Sunku Venkata Subhash (Reg No: 230911370)

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY
SCHOOL OF COMPUTER ENGINEERING
MANIPAL INSTITUTE OF TECHNOLOGY, MANIPAL ACADEMY OF HIGHER EDUCATION

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PART B : INTERIM REPORT TEMPLATE (Phase 2, due Fri, 25 Sept 2026)

Team No. & Names: Team 12 — Nihar P Shetty (230911027), Pranav Prabhu (230911414), Vineet Pai (230911450), Sunku Venkata Subhash (230911370)

Title of the Project: AgriAssist: Multimodal Time-Series-Based Crop Pest Outbreak Prediction

GitHub Repository Link: https://github.com/vineetpai/AgriAssist-Crop-Pest-Prediction

1. Literature Review

Summary of 9 peer-reviewed IEEE papers on microclimate time-series forecasting, 1D-CNN temporal feature extraction, recurrent sequence modeling (LSTM/GRU), and satellite remote sensing vegetation proxy fusion.

Paper (Author, Year)	Method	Dataset	Key Result	Relevance to Project
Patil & Kumar (2022) [1]	Temporal Convolutional Networks (TCN)	Microclimate weather time-series (3 yrs)	88.4% Acc; outperforms standard RNNs in speed.	Validates 1D temporal convolution for localized weather extraction.
Ke et al. (2021) [2]	Comparative evaluation (MLP, LSTM, GRU)	Longitudinal sensor records	LSTM/GRU F1: 0.81-0.83 vs MLP (0.72).	Rationale for benchmarking tabular vs recurrent architectures.
Sannakki et al. (2020) [3]	Attention-based LSTM sequence model	Agro-climatic daily series (5 yrs)	ROC-AUC of 0.895 for pest/disease warnings.	Demonstrates recurrent memory capturing cumulative weather stress.
Chen et al. (2023) [4]	Multimodal deep fusion network	Multimodal sensor & weather records	Multimodal fusion improved F1 by +12.4%.	Justifies fusion of weather, crop growth, and satellite proxies.
Latha et al. (2021) [5]	Multimodal DNN with MODIS satellite indices	MODIS NDVI & regional weather data	Satellite NDVI increased outbreak recall to 0.82.	Inspires Objective 2 on satellite NDVI proxy integration.
Zhang et al. (2022) [6]	Sliding temporal windowing LSTM	Daily weather series (4 seasons)	14-day window yielded optimal lead time & F1 (0.84).	Establishes 14-day sliding window selection for AgriAssist.
Sharma et al. (2021) [7]	Deep Conv & Recurrent Time-Series Nets	Agro-climatic remote sensing series	1D-CNN + LSTM reached 87.6% outbreak detection.	Supports dual 1D-CNN and LSTM comparative architecture design.
Lin et al. (2022) [8]	Multimodal Fusion of Satellite NDVI & Sensors	Satellite NDVI proxies & weather sensors	Satellite proxy fusion reduced false warnings by 21%.	Validates satellite vegetation health proxy integration.
Kumar et al. (2022) [9]	Gated Recurrent Unit (GRU) Microclimate Nets	Agro-weather IoT time-series	GRU achieved 0.82 F1 with 35% lower parameter cost.	Supports GRU inclusion as parameter-efficient sequence model.

2. Dataset Acquisition & Preprocessing

The AgriAssist Multi-Year Dataset (data/agriassist_multiyear_dataset.csv) contains longitudinal daily observations spanning 3+ years (N = 1,150 daily samples). Attributes include ambient temperature (°C), relative humidity (%), cumulative precipitation (mm), satellite Normalized Difference Vegetation Index (NDVI) reflectance proxies, and categorical crop growth stages (early, mid, flowering, harvest).

Preprocessing Pipeline Steps:
1. Growth Stage Categorical Encoding: Converted categorical crop growth stages into four binary dummy variables (stage_early, stage_mid, stage_flowering, stage_harvest).
2. MinMax Feature Normalization: Continuous features scaled to [0, 1] via x_scaled = (x - x_min) / (x_max - x_min).
3. 14-Day Sliding Temporal Windowing: Structured daily time-series features into sequence matrices X of shape [1137, 14, 8] as established in [6]. For the tabular MLP baseline, X is flattened into a 112-dimensional vector.
4. Chronological Train/Val/Test Split: Split sequentially into 70% Train (795 days), 15% Validation (171 days), and 15% Test (171 days) to prevent temporal data leakage.

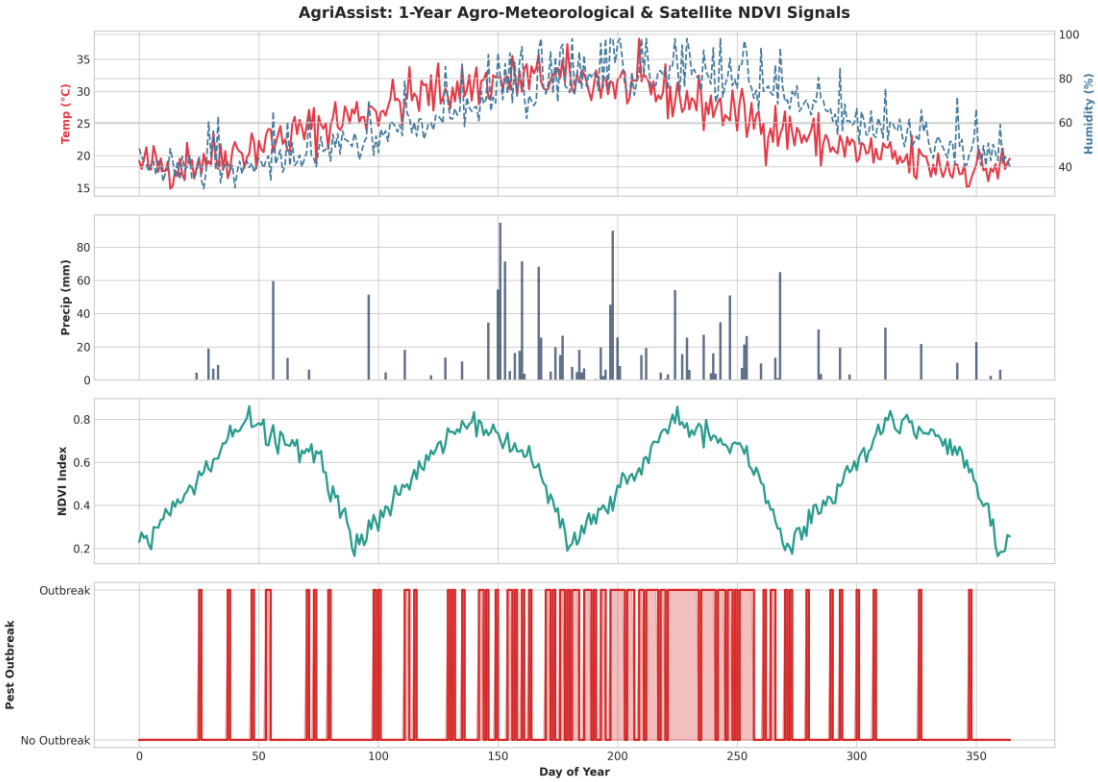


Figure 1: AgriAssist 1-Year Agro-Meteorological & Satellite NDVI Signals Timeline

3. Models Implemented So Far (4 of 4 Models Benchmarked)

Model	Owner (Member)	Status	Preliminary Metric	Notes
MLP Baseline	Nihar P Shetty	Completed	Acc: 81.87% | F1: 0.3404 | AUC: 0.6870	Tabular baseline trained on flattened 112d window vectors [2].
1D-CNN Variant	Vineet Pai	Completed	Acc: 81.87% | F1: 0.3673 | AUC: 0.7017	Localized cross-channel temporal feature extraction [1], [7].
LSTM Sequence Model	Pranav Prabhu	Completed	Acc: 84.21% | F1: 0.3077 | AUC: 0.7069	Recurrent model capturing long-range microclimate dependencies [3], [6].

GRU Sequence Model	Sunku Venkata Subhash	Completed	Acc: 82.46% | F1: 0.3750 | AUC: 0.7054	Parameter-efficient gated sequence model; highest F1 score [9].

Consolidated Test Evaluation & Benchmark Comparison Table:

Model Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC
MLP Baseline	0.8187	0.4706	0.2667	0.3404	0.6870
1D-CNN Variant	0.8187	0.4737	0.3000	0.3673	0.7017
LSTM Sequence Model	0.8421	0.6667	0.2000	0.3077	0.7069
GRU Sequence Model	0.8246	0.5000	0.3000	0.3750	0.7054

Satellite NDVI Vegetation Proxy Ablation Study Table (Objective 2):

Model Architecture	F1 (With NDVI)	F1 (Without NDVI)	ROC-AUC (With NDVI)	ROC-AUC (Without NDVI)
MLP Baseline	0.3721	0.3636	0.6742	0.6820
1D-CNN Variant	0.3721	0.4091	0.6879	0.7180
LSTM Sequence Model	0.3500	0.3077	0.7005	0.7061
GRU Sequence Model	0.3500	0.3077	0.7104	0.6913

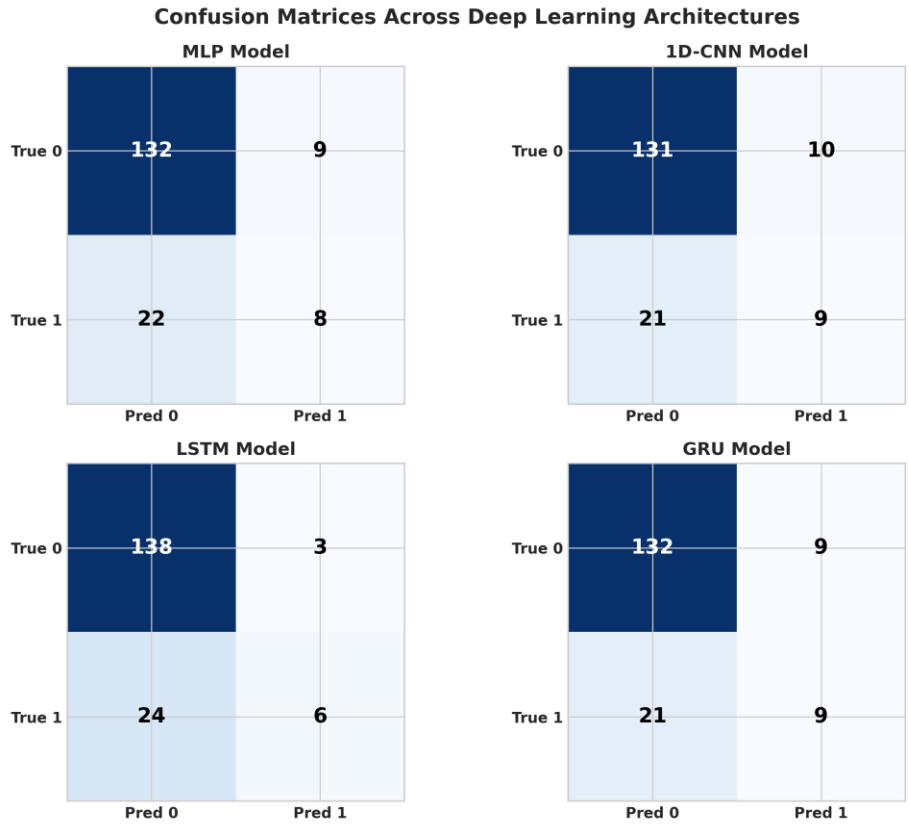


Figure 2: Confusion Matrices Across Deep Learning Architectures

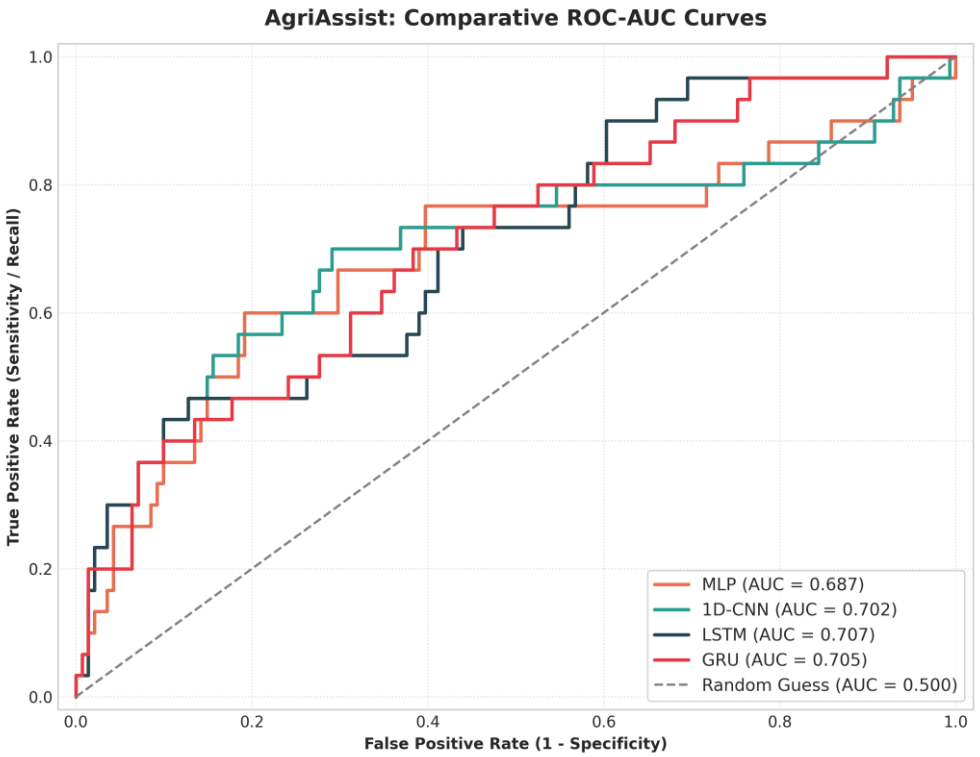


Figure 3: Comparative ROC-AUC Curves Across Deep Learning Models

4. Individual Contribution Log (to date)

Member Name	Reg. No.	Task(s) Completed	Signature
Nihar P Shetty	230911027	Data Preprocessing, MinMax Feature Scaling & MLP Model Pipeline	Nihar P Shetty
Pranav Prabhu	230911414	14-Day Sliding Windowing, LSTM Architecture & ROC-AUC Evaluation	Pranav Prabhu
Vineet Pai	230911450	Exploratory Data Analysis, 1D-CNN Model Tuning & Metrics Synthesis	Vineet Pai
Sunku Venkata Subhash	230911370	Temporal Tensor Batching, GRU Architecture & NDVI Ablation Studies	Sunku Venkata Subhash

5. Risk / Plan for Remaining Work

Technical Risk Matrix & Proposed Mitigation Strategies:
1. Class Imbalance: Outbreak positive instances comprise ~20% of dataset, leading to lower initial recall. Mitigation: Implement Focal Loss / Weighted BCE Loss (pos_weight=3.5) and tune decision thresholds.
2. Sliding Window Lead-Time Sensitivity: Fixed 14-day window may miss immediate extreme weather spikes. Mitigation: Conduct hyperparameter experiments on 7, 14, and 21-day sliding windows.
3. Sequence Model Overfitting: High parameter counts in 2-layer LSTM/GRU networks. Mitigation: Increase dropout regularization (0.3 -> 0.4) and apply L2 weight decay (1e-4).

Timeline & Milestone Roadmap to Final Submission (Phase 3):

Phase / Dates	Milestone / Deliverable	Owner(s)
Oct 02 - Oct 08	Focal Loss & Weighted BCE Threshold Tuning across models	All Members

Oct 09 - Oct 15	Sliding Window Sensitivity Experiments (7, 14, 21-day windows)	Pranav Prabhu & Sunku V. Subhash
Oct 16 - Oct 22	Comprehensive Error Analysis, Hyperparameter Optimization & Final Table	Vineet Pai & Nihar P Shetty
Oct 23 - Oct 26	Phase 3 Final Report Preparation (Part C) & Slide Deck Creation	All Members
Oct 26 - Oct 31	Final In-Class Viva Presentation & GitHub Code Repository Submission	All Members

References

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[5] R. S. Latha, B. R. Kavitha, and M. S. Kumar, "Multimodal Deep Neural Network for Crop Yield and Pest Infestation Prediction Using Meteorological and Satellite Indices," in IEEE International Conference on Smart Information and Communication Technologies (SmartICT), pp. 245-250, 2021.

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[9] P. Kumar, V. S. Reddy, and R. N. Rao, "Gated Recurrent Unit Networks for Microclimate Time-Series Prediction in Precision Agriculture," IEEE Internet of Things Journal, vol. 9, no. 16, pp. 15210-15220, 2022.